

From Statistical Experiment to the Metre

Stratum 7 of the Theorem Ladder: a Non-Circular Physical-Distance Identification, its Established Imports, and a 20-Check Hostile Battery

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September 28, 2026

Abstract

The distance-and-angle rung characterizes RG distance d_{RG} as the intrinsic length of the nuisance-efficient Fisher tensor on a regular quotient stratum. This note builds the missing step to *physical* distance and attacks every joint before a referee can. The central negative result is that d_{RG} is not, and no function of it can be, a physical distance: it scales as $\sqrt{N}/\sigma$ and contracts under garbling while displacement does not (PD01–PD02). What survives is structure, not magnitude. If displacements are realized as experiment automorphisms whose composition is concatenation, Cauchy–Hölder forces a ratio scale; if a rotation group acts irreducibly, Schur forces every invariant metric to be $\lambda_E \delta$; so the Fisher metric equals an experiment-dependent constant times one experiment-independent metric h . No statistical, thermodynamic or conal datum fixes the constant: the metre lies in an exact dilation fibre (PD05), Čencov leaves one free constant (PD18), thermodynamic length is unit-blind (PD17), and $\{k_B, T, m\}$ contains no length (PD15). The minimal non-circular anchor is two-way radar with c and $\Delta\nu_{\text{Cs}}$ exact by the 2019 SI definition; the factor $\frac{1}{2}$ is forced by the Reichenbach ε -gauge (PD07). Radar distance equals rod distance only where exchanged clock frequencies agree, i.e. where the scale cocycle is exact (PD13–PD14, rung R23). The deterministic battery (seed 20260928) passes 20/20 registered checks, falsifies 11 naive claims with explicit counterexamples, survives three further seeds, and a mutation canary confirms that each of four sabotaged structures is caught.

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1 The question, typed

The ladder [1] is conformal-complete and scale-open by construction: the cone yields $[g]$ (R18), a single worldline underdetermines the conformal factor (R22), and a metric representative exists iff the scale cocycle is exact (R23). The distance paper [2] adds d_{RG} on regular strata. The question is therefore sharply posed: *which additional, independently testable structures convert d_{RG} into metres, and can this be done without importing a length?* We use the ladder tags: forced, imported, gated, declared, no-go, open.

2 Negative stratum: what d_{RG} is not

No-Go 2.1 (Experiment relativity, PD01). For a Gaussian location experiment with N samples and noise σ , $d_{\text{RG}} = \sqrt{N}|\Delta x|/\sigma$. Over $N \in [1, 10^4]$, $\sigma \in [0.01, 1]$ the same $\Delta x = 1$ m yields d_{RG} spanning a factor 10^4 ; $(N, \sigma, \Delta x) = (4, 1, 1)$ and $(1, 1, 2)$ give the same $d_{\text{RG}} = 2$.

No-Go 2.2 (Garbling no-go, PD02). Let $\mathcal{C}$ be closed under additive-noise garbling. There is no function F with $F(d_{\text{RG}}) = d_{\text{phys}}$ on $\mathcal{C}$. *Proof.* $d_{\text{RG}}(\Delta x, \tau) = \Delta x/\sqrt{\sigma^2 + \tau^2}$ is strictly decreasing to 0 in τ ; $d_{\text{RG}}(1, 0) = d_{\text{RG}}(2, \sqrt{3}\sigma)$ while d_{phys} differs. (Recovered $\tau^* = 1.7320508$.) $\square$

Remark 2.3. This does not contradict [2]: monotonicity under garbling is the *correct* property of a distinguishability length and the *wrong* property of a physical length. The two objects must be separated before any identification.

Proposition 2.4 (Intrinsic versus chordal, PD04). *Only the intrinsic length is additive. The Bhattacharyya angle $2 \arccos \text{BC}(a)$ between p and $p(\cdot - a)$ gives ratios $\text{ang}(2a)/\text{ang}(a) = 1.877$ (Gaussian), 1.738 (Laplace), 1.937 (logistic), 1.831 (Student-3), 1.751 (Cauchy) instead of 2, and saturates at π . Physical distance may be read only from intrinsic length or from concatenation of small steps.*

Proposition 2.5 (Nuisance wall, PD09). *g_{eff} is invariant under $\nu \mapsto \nu + f(q)$ (defect 2.9×10^{-14}) but depends on the declared target chart $q \mapsto q + b\nu$. For $I_\varepsilon = \begin{pmatrix} a & \rho\sqrt{a\varepsilon} \\ \rho\sqrt{a\varepsilon} & \varepsilon \end{pmatrix}$, $\lim_{\varepsilon \rightarrow 0+} g_{\text{eff}} = a(1 - \rho^2) = 0.72$ while $g_{\text{eff}}(0) = a = 2$: the pseudoinverse Schur complement is discontinuous at a nuisance rank wall. A declared nuisance bound (prior precision 1) restores continuity ($1.9999999999987 \rightarrow 2$).*

This is an omitted step in [2]: the “regular stratum” hypothesis must include *nuisance* rank constancy, or a declared nuisance bound, not only target rank constancy.

3 Positive stratum: what survives is structure

Definition 3.1 (Operational displacement). A displacement is a physical operation T on the source (or apparatus) such that the experiment family is covariant: $P_{T \cdot \theta} = T_* P_\theta$. Covariance is an equality-of-law statement and is testable with the R1 primitive. Concatenation of displacements is composition of operations.

Lemma 3.2 (Translation covariance forces a constant metric, PD03). *For a regular family $p(y - \theta)$ the Fisher metric is θ -independent. Monte-Carlo score variance at $\theta \in \{-3, 0, 2.5, 7\}$ (4×10^5 samples) is constant within 0.37% for Gaussian, logistic and Cauchy families; numerical Fisher constants match $1, 1, \frac{1}{3}, \frac{2}{3}, \frac{1}{2}$ to 5×10^{-11} . Control: the non-covariant Poisson-rate family assigns lengths 0.828, 0.636, 0.309 to equal steps.*

Theorem 3.3 (Ratio scale; imported: Cauchy, Hölder, KLST [4, 5]; PD12). *A monotone representation ψ of concatenation with $\psi(a \circ b) = \psi(a) + \psi(b)$ is linear, unique up to one positive constant. Cubic, $\sinh$ and $x + 0.1 \sin x$ rulers violate additivity by 44.9, 18.1, 0.25.*

Theorem 3.4 (Invariant-metric uniqueness; imported: Schur; PD10). *Let a compact group K act linearly on the displacement space V . The K -invariant symmetric bilinear forms are one-dimensional iff K acts irreducibly over $\mathbb{R}$ (the commutant is $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$, whose symmetric part is $\mathbb{R}$). Hence, under irreducible rotation covariance, every invariant metric is $\lambda\delta$. Computed dimensions: translations only 3 (2D) and 6 (3D); C_2 : 3; C_4 in 2D: 1; C_4 about an axis in 3D: 2; tetrahedral $\langle C_3, D \rangle$: 1. Anisotropic noise $\text{diag}(1, 9)$ maps physically orthogonal displacements to an RG angle of 141.7° .*

No-Go 3.5 (Flat is not physical, PD19). For the response $\mu(x, y) = (x + ay^2, y)$ the Fisher metric has Gaussian curvature exactly 0 (symbolic Brioschi), yet equal physical steps receive RG lengths 1.077 and 1.562. “ $\bar{g} = \lambda^2 I$ on some chart” in [2] must read “on the chart supplied by the covariance group”.

Remark 3.6 (Helmholtz–Lie and Lambert; imported, partly open). Free mobility of rigid operations forces constant curvature [6]. In the flat case the automorphism group contains dilations and there is no intrinsic unit; in the spherical/hyperbolic case the curvature radius is an absolute unit (Lambert). Which case holds is decided by an angle-sum test, which is an RG angle measurement. This is the only place where geometry could, in principle, carry its own unit.

4 No internal unit

No-Go 4.1 (Dilation fibre, PD05). If readout is in instrument units $u = y/s$ with s unanchored, the law depends on $(x/s, \sigma/s)$; the Fisher matrix in (x, σ, s) has rank 2 at all 200 random points and the dilation generator (x, σ, s) has quadratic form 0. The unit is an arrow of the observational kernel groupoid.

No-Go 4.2 (Čencov constant, PD18; imported [3]). Markov-invariant metrics are c -Fisher, $c > 0$ free (embedding defect 1.1×10^{-13} ; Euclidean metric on the simplex fails with defect 1.52).

No-Go 4.3 (Thermodynamic insufficiency, PD15, PD17; Buckingham [8]). Thermodynamic length $\int \sqrt{\text{Var} E} d\beta$ is invariant under $E \mapsto aE$, $\beta \mapsto \beta/a$ (deviation 0 for $a \in \{10^{-3}, 0.5, 7, 10^4\}$). Over (M, L, T, Θ) , $\{k_B, T, m\}$, $\{\nu\}$ and $\{h, k_B, T\}$ admit no length; $\{c, \nu\} \rightarrow c/\nu$, $\{h, m, k_B, T\} \rightarrow h(mk_B T)^{-1/2}$, $\{k_B, T, m, g\} \rightarrow k_B T/(mg)$. A minimal thermodynamic calibration fixes an *energy* scale; a length needs an action or a clock–speed pair. Thermodynamic length [9] is a dissipation bound, not a ruler.

5 The anchor

Gate 5.1 (SI anchor; declared). $c = 299\,792\,458 \text{ ms}^{-1}$ and $\Delta\nu_{\text{Cs}} = 9\,192\,631\,770 \text{ Hz}$ are exact by definition [10]. The length $c/\Delta\nu_{\text{Cs}} = 0.0326122557 \text{ m}$ is therefore not measured with any rod.

Proposition 5.2 (Anchor breaks the dilation, PD06). *Two-way counts $k \sim \mathcal{N}(2d\nu/v, \sigma_k^2)$ give Fisher rank 1 in (d, v, ν) . With $v = c$, $\nu = \Delta\nu_{\text{Cs}}$ fixed, $I_{dd} = 41788.4$ at $\sigma_k = 0.3$. In a medium with unanchored v the (d, v) dilation persists (form 0).*

Proposition 5.3 (Two-way factor forced; PD07; Reichenbach [11]). *With one-way speeds $c/(2\varepsilon)$ and $c/(2(1 - \varepsilon))$, the round-trip time is ε -independent (spread 2×10^{-15} over 500 values) and carries zero Fisher information on ε . Thus $d_{\text{rad}} = c\tau/2$: the $\frac{1}{2}$ is forced by the synchronization gauge, not fitted.*

Remark 5.4 (Reading of $d = \Theta\gamma M_1$). In the radar branch one admissible reading of the rank-one law of [2] is $\Theta = c$ (exact), $\gamma = \frac{1}{2}$ (forced), $M_1 = \tau$ (clock proper time of the round trip). It is legitimate only under the gates below; in a medium, $\Theta = v$ must itself be anchored (PD06, PD08).

Proposition 5.5 (Traceability, PD16). *Anchor routes are acyclic iff their dependency graph terminates in SI-defined constants and counts: radar ($c, \Delta\nu_{Cs}$) and thermal de Broglie (h, k_B , Johnson-noise T , atom mass via h/m) are acyclic; equipartition with a measured stiffness, barometric height with measured g , and sound ranging with measured v_s re-import the metre.*

6 Gates between radar distance and metric distance

Proposition 6.1 (Cone plus one clock fixes radar distance only, PD13). *In $ds^2 = W(x)^2(-dt^2 + dx^2)$ with a clock at $W(0) = 1$, null rays and hence radar distance are W -independent (spread 0 over 300 factors), while rod distance $\int_0^X W$ differs by up to 4.44. The exchanged-frequency ratio $W(X)/W(0)$ is the observable that detects the gap. This refines R22 and is consistent with the cone theorem [13]: a single worldline fixes radar distance but not the conformal factor elsewhere.*

Proposition 6.2 (Scale holonomy, PD14; R23; EPS [12]). *Transporting the unit with a Weyl form φ multiplies it by $\exp \oint \varphi$: 1.0000000 for exact φ , 3.5135838 versus predicted $e^{0.4\pi} = 3.5135856$ for $d\varphi = 0.4 dx \wedge dy$ on the unit disc. A global metre exists iff there is no second-clock effect.*

Proposition 6.3 (Heterogeneous media, PD08). *Delays identify integrated slowness only: (10 m, 1500), (20 m, 3000) and a two-layer profile share $T = 6.667$ ms. On a 4×4 grid straight-ray nullity is 12 (one family), 9 (two), 1 (with diagonals): path diversity shrinks but need not remove the invisible slowness sector (cf. boundary rigidity [14]).*

Proposition 6.4 (One anchor: rank matters, PD11; Beckman–Quarles [7]). *$\phi(x) = x + 0.1 \sin 2\pi x$ preserves every unit distance (7×10^{-15}) but not half-units (0.20). The same map in 2D breaks oblique unit distances (0.12). In rank ≥ 2 a unit-distance-preserving map of Euclidean space is an isometry; in rank one, additivity (Thm. on ratio scale) is additionally required.*

Proposition 6.5 (Held-out anchor, PD20). *With one radar anchor at 1 m, a covariant experiment predicts a held-out 15 m anchor exactly; a response $x + 0.02x^2$ passes the first anchor by construction and misses the second by 27.5%. A single anchor is never self-validating.*

7 The identification theorem

Theorem 7.1 (Physical-distance identification). *Let $\{\mathcal{E}\}$ be a family of regular classical RG experiments on a connected displacement chart U , and assume:*

- G1 **Covariance**: a Lie group G of physical operations acts transitively on U , the family is G -covariant in the sense of Def. 3.1, and the isotropy acts irreducibly (tested by equality of laws);*
- G2 **Regularity**: constant target and nuisance rank, or a declared nuisance bound;*
- G3 **Anchor**: two-way radar in Cs cycles with $c, \Delta\nu_{Cs}$ exact, the radar protocol transported by the same G ;*
- G4 **Scale integrability**: exchanged-frequency ratios are trivial on U (exact scale cocycle);*
- G5 **Medium**: propagation speed is c or is itself anchored by an acyclic route;*
- G6 **Held-out validation**: a second anchor is predicted within a declared tolerance.*

Then there is a unique G -invariant Riemannian metric h on U whose distance equals $d_{\text{rad}} = (c/2\Delta\nu_{\text{Cs}})n_{\text{Cs}}$, and for every $\mathcal{E}$ in the family

$$g_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h, \quad d_{\text{phys}} = d_{\text{RG}}^{\mathcal{E}}/\lambda_{\mathcal{E}},$$

with $\lambda_{\mathcal{E}} > 0$ constant on U , fixed by one anchor and validated by $G6$. RG angles equal physical angles for every $\mathcal{E}$.

Proof sketch. $G1$ and naturality of Fisher information under experiment isomorphism [2] make $g_{\mathcal{E}}$ G -invariant; by Theorem 3.4 it is $\lambda_{\mathcal{E}}^2 \delta$ in the group-adapted chart (not an arbitrary flat chart, PD19). $G3$ – $G4$ make d_{rad} G -invariant and additive along concatenations; by the ratio-scale theorem and Theorem 3.4 again, d_{rad} is the distance of a unique invariant metric h , so $g_{\mathcal{E}} \propto h$. $G5$ excludes the (d, v) fibre, $G2$ excludes the nuisance wall, and $G6$ excludes covariance-breaking responses that fit one anchor. $\square$

Corollary 7.2 (Minimal calibration). *The minimal non-circular calibration is one SI-defined clock-speed pair entering through two-way timing, plus one held-out anchor. Nothing thermodynamic, conal, or statistical can replace the pair (No-Go 4.1, PD15, PD17, PD18).*

Each hypothesis is necessary: removing $G1$ gives PD10/PD19, $G2$ gives PD09, $G3$ gives PD05, $G4$ gives PD13/PD14, $G5$ gives PD06/PD08, $G6$ gives PD20.

8 Ladder insertion (Stratum 7)

Rung	Statement	Type	Depends / attack
R30	d_{RG} dimensionless, chart invariant; unique up to Čencov constant	imported	R8, PD18
R31	d_{RG} not physical; no $F(d_{\text{RG}}) = d_{\text{phys}}$ on garbling-closed classes	no-go	R30, PD01–02
R32	Only intrinsic length additive; chordal divergences are not	gated	R30, PD04
R33	g_{eff} target-chart dependent; discontinuous at nuisance rank wall	gated	R11, PD09
R34	Displacement = experiment automorphism; covariance testable by R1	declared	R1, R5
R35	Concatenation $\Rightarrow$ ratio scale (Cauchy–Hölder)	imported	R34, PD12
R36	Irreducible isotropy $\Rightarrow$ invariant metric $\lambda\delta$ (Schur); flat $\neq$ physical	imported (gated)	R34, PD03,10,19
R37	Flat case has no intrinsic unit; curved case has Lambert unit	imported / open	R36
R38	Unit lies in an exact dilation fibre; thermodynamics has no length	no-go	R2, PD05,15,17
R39	SI anchor: two-way radar, $c, \Delta\nu_{\text{Cs}}$ exact; $\gamma = \frac{1}{2}$ forced	declared (anchor)	R18, R22, PD06,07,16
R40	Radar = rod distance iff exact scale cocycle (no redshift)	gated	R23, PD13,14
R41	Media: speed anchor or tomography; blind slowness sector	gated	R12, PD08

Rung	Statement	Type	Depends / attack
R42	Rank ≥ 2 : one unit anchor suffices (Beckman–Quarles); rank one needs R35	imported (gated)	PD11
R43	Identification theorem with held-out anchor	gated	R30–R42, PD20
R44	Stationary/non-static GR, quantum Fisher, non-dominated experiments	open	—

9 Corrections forced on the distance–angle paper

1. “Regular stratum” must include nuisance rank constancy or a declared nuisance bound (PD09).
2. “ $\bar{g} = \lambda^2 I$ on a chart” must name the chart supplied by the covariance group (PD19).
3. “Independent scale calibration setting $\lambda = 1$ ” must be per experiment, since λ scales as $\sqrt{N}/\sigma$ (PD01), and must be held-out validated (PD20).
4. The rank-one reduction $d = \Theta_\gamma M_1$ needs an additivity law; one anchor does not calibrate a rank-one channel (PD11).
5. Distinguishability language must be restricted to intrinsic length (PD04).

10 Reproducibility ledger

Run	Result	Note
seed 20260928	20/20 pass, 11 naive claims falsified	primary
seeds 1, 42, 777	20/20 pass each	seed robustness
--canary	4/4 mutations caught	PD10, PD14, PD15, PD20 sabotaged; only target fails
self-attack	NaN-silent pass found and removed	logistic overflow; non-finite evidence now fails
self-attack	tautological ratio check replaced	Monte-Carlo score variance + Poisson control

11 Non-claims

Non-claim 11.1. No physical experiment is validated here; all gates are simulated or analytic.

Non-claim 11.2. The SI anchor is a declared convention, not derived; RG shows only that some such anchor is necessary and that this one is sufficient and acyclic under G1–G6.

Non-claim 11.3. Curved, stationary and non-static spacetimes (Shapiro delay, frame dragging) and quantum Fisher geometry are outside R43.

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